Assignment: Build a Simple Coding Agent

Build a **Claude Code–like coding agent using plain Python**. Do not use LangChain, LangGraph, CrewAI, or other agent frameworks.

Expectations

Your agent should follow a basic **ReAct loop**:

Reason Act Observe Repeat Final Answer

It should be able to:

- Read files
- Create new files
- Edit existing files
- Delete files
- List files/directories
- Execute terminal commands
- Return tool outputs back to the LLM as observations
- Perform multiple tool calls for a single user request

For now, focus mainly on **code generation and file operations**, rather than debugging existing projects.

Example prompt:

```
Create a simple Python Todo application.

Requirements:
- Create a main.py file
- Allow users to add and list todos
- Store todos in a JSON file
- Create any additional files you think are required
- Run the application once after creating it
```

A possible agent flow could be:

```
Understand requirements
  Check existing files
  Create required files
  Write the code
  Run the program
  Observe the output
  Return final response
```

Human-in-the-Loop

Add basic approval before executing potentially dangerous commands such as:

```
rm -rf
git reset --hard
git push
```

Also restrict file access to a specific project/workspace directory.

Minimum Requirements

- Plain Python agent loop
- Any LLM API/SDK
- Filesystem CRUD tools
- Terminal execution tool
- Multi-step tool calling
- Basic security checks
- Human approval for dangerous actions
- Maximum step limit to avoid infinite loops